

SUMIT KUMAR

PhD Candidate, Centre for Atmospheric & Oceanic Sciences (CAOS), Indian Institute of Science,
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RESEARCH PROFILE

Atmospheric and climate scientist studying the radiative forcing and feedback mechanisms of key anthropogenic climate drivers — methane, black carbon aerosol, and stratospheric sulfate aerosol — using the radiative kernel technique and Earth System Model (CESM) experiments. Experienced in designing, running, and diagnosing multi-decadal coupled climate simulations, with particular interest in comparing greenhouse gas, aerosol, and solar geoengineering (SRM) forcings and their distinct climate responses.

EDUCATION

Ph.D., Atmospheric & Oceanic Sciences — IISc, Bengaluru	Feb 2021 – 2026 (expected)
M.Sc., Physics — S.G.T.B. Khalsa College, University of Delhi	2017 – 2019
B.Sc. (Hons.), Physics — Kirori Mal College, University of Delhi	2014 – 2017

RESEARCH EXPERIENCE

PhD Researcher — Centre for Atmospheric & Oceanic Sciences (CAOS), IISc Bengaluru

(Feb 2021 – Present)

- Methane vs. CO₂ radiative efficacy:** applied the radiative kernel technique to isolate and contrast the slow, surface-temperature-mediated climate response to CH₄ and CO₂ forcing, identifying key differences in forcing efficacy and feedback contributions (Journal of Climate, 2025; presented at EGU 2025).
- Black carbon aerosol forcing:** extended the radiative kernel and feedback-decomposition framework to black carbon aerosol perturbations, showing weaker climate sensitivity to BC than to CO₂ forcing (submitted to Journal of Climate, June 2026).
- Stratospheric aerosol injection & solar geoengineering:** designed and ran multi-decadal CESM1.2.2 (CAM5 physics, Bulk Aerosol Model) experiments injecting volcanic sulfate aerosol at multiple stratospheric altitudes under a doubled-CO₂ background, using both slab-ocean and prescribed-SST configurations, to contrast the climate responses of sulfate-aerosol and solar-constant-reduction SRM strategies (manuscript in preparation).
- Model development & diagnostics:** diagnosed and resolved configuration issues at the Fortran source-code level within CESM/CAM (physics-buffer interfaces, prescribed-aerosol subroutines), and ported MATLAB-based radiative-kernel diagnostics to Python for CAM5 feedback analysis.

PUBLICATIONS

Peer-Reviewed

1. **Kumar, S.,** Seshadri, A. K., & Bala, G. (2025). Contrast in slow climate response to methane and carbon dioxide forcings identified using the radiative kernel technique. *Journal of Climate*. <https://doi.org/10.1175/JCLI-D-24-0728.1>

Under Review

2. **Kumar, S.,** Seshadri, A. K., & Bala, G. Radiative Kernel Analysis Reveals Weaker Climate Sensitivity to Black Carbon Forcing than to Carbon Dioxide. *Journal of Climate* (submitted June 2026; under review).

In Preparation

3. **Kumar, S.,** Seshadri, A. K., & Bala, G. Contrast in Solar and Sulfate Aerosol Forcings Does Not Translate into Their Climate Responses. (manuscript in preparation).

CONFERENCE PRESENTATIONS

Kumar, S. (2025). An Estimation of the Efficacy of Methane Radiative Forcing Using Radiative Kernels. European Geosciences Union (EGU) General Assembly, Vienna, Austria. <https://doi.org/10.5194/egusphere-egu25-5950>

AWARDS & SCHOLARSHIPS

- MHRD Fellowship, Indian Institute of Science
- Grantham Scholarship
- GATE (Physics) 2020 — All India Rank 386

TECHNICAL SKILLS

- **Climate / Earth System Models:** CESM (CAM5 physics, Bulk Aerosol Model) — source-code-level experience
- **Programming & Scripting:** Python (xarray, NumPy, NetCDF), MATLAB, Fortran, Bash/Shell
- **Computing Environment:** Linux/Unix, high-performance computing (HPC) clusters
- **Core Techniques:** Radiative kernel technique, radiative forcing & feedback analysis, aerosol–climate interactions
- **Scientific Visualization:** Blender (3D scientific illustration)

PROFESSIONAL EXPERIENCE

Subject Matter Expert (Physics), Chegg India Pvt. Ltd.

May 2019 – Sep 2020